Precipitation and Net Ionic Equations

Answer key with worked solutions

A. Solubility

1. Soluble or insoluble.

> (a) **insoluble** (b) **soluble** (c) **insoluble** (d) **soluble** (e) **insoluble** (f) **soluble**

B. Predicting precipitates

2. Precipitates.

> (a) **Yes**: calcium carbonate, $\mathrm{CaCO_3}$ (white). (b) **No**: $\mathrm{KNO_3}$ and NaOH are both soluble. (c) **Yes**: copper(II) hydroxide, $\mathrm{Cu(OH)_2}$ (blue). (d) **Yes**: silver iodide, AgI (yellow).

C. Writing equations

3. Barium chloride and sodium sulfate.

> (a) $\mathrm{BaCl_2\,(aq) + Na_2SO_4\,(aq) \longrightarrow BaSO_4\,(s) + 2\,NaCl\,(aq)}$
>
> (b)
>
> $$\mathrm{Ba^{2+}\,(aq) + 2\,Cl^-\,(aq) + 2\,Na^+\,(aq) + {SO_4}^{2-}\,(aq)} \longrightarrow \mathrm{BaSO_4\,(s) + 2\,Na^+\,(aq) + 2\,Cl^-\,(aq)}$$
>
> (c) $\mathrm{Ba^{2+}\,(aq) + {SO_4}^{2-}\,(aq) \longrightarrow BaSO_4\,(s)}$
>
> (d) $\mathrm{Na^+}$ and $\mathrm{Cl^-}$

4. Net ionic equations.

> (a) $\mathrm{Ca^{2+}\,(aq) + {CO_3}^{2-}\,(aq) \longrightarrow CaCO_3\,(s)}$
>
> (c) $\mathrm{Cu^{2+}\,(aq) + 2\,OH^-\,(aq) \longrightarrow Cu(OH)_2\,(s)}$
>
> (d) $\mathrm{Ag^+\,(aq) + I^-\,(aq) \longrightarrow AgI\,(s)}$

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5. Neutralization.

> $\mathrm{H^+(aq) + OH^-(aq) \longrightarrow H_2O(l)}$. Strong acids and strong bases are fully ionized, so the anion of the acid and the cation of the base are always **spectators**: the only real change is $\mathrm{H^+ + OH^-}$ forming water.

D. Calculations

6. Sulfate concentration.

> $$n(\mathrm{BaSO_4}) = \frac{0.466\ \mathrm{g}}{233.39\ \mathrm{g/mol}} = 1.997 \times 10^{-3}\ \mathrm{mol}$$
>
> $1\ \mathrm{mol\ SO_4^{2-}}$ gives $1\ \mathrm{mol\ BaSO_4}$, so $n(\mathrm{SO_4^{2-}}) = 1.997 \times 10^{-3}\ \mathrm{mol}$.
>
> $$c = \frac{1.997 \times 10^{-3}\ \mathrm{mol}}{0.0500\ \mathrm{L}} = \mathbf{0.0399\ mol/L}$$

7. Lead(II) iodide.

> $n(\mathrm{I^-}) = 0.120\ \mathrm{mol/L} \times 0.0200\ \mathrm{L} = 2.40 \times 10^{-3}\ \mathrm{mol}$
>
> $\mathrm{Pb^{2+} + 2\,I^- \longrightarrow PbI_2}$: $2\ \mathrm{mol\ I^-}$ give $1\ \mathrm{mol\ PbI_2}$, so $n(\mathrm{PbI_2}) = 1.20 \times 10^{-3}\ \mathrm{mol}$.
>
> $$m = 1.20 \times 10^{-3}\ \cancel{\mathrm{mol}} \times 461.0\ \frac{\mathrm{g}}{\cancel{\mathrm{mol}}} = \mathbf{0.553\ g}$$

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E. Challenge

8. Purity of sodium chloride.

(a)

$$m(\text{Cl}) = 0.717\ \text{g AgCl} \times \frac{35.45\ \text{g Cl}}{143.32\ \text{g AgCl}}$$
$$= \mathbf{0.1773\ g}$$

(b)

$$\%\text{Cl} = \frac{0.1773\ \text{g}}{0.300\ \text{g}} \times 100\%$$
$$= \mathbf{59.1\%}$$

(c) Pure NaCl is $\frac{35.45\ \text{g/mol}}{58.44\ \text{g/mol}} \times 100\% = 60.66\%$ chlorine.

$$\text{purity} = \frac{59.12\%}{60.66\%} \times 100\%$$
$$= \mathbf{97.5\%}$$

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